

AIML SESSION 01 ASSIGNMENT SUBMISSION

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Domain/Subject : AIML (Artificial Intelligence & Machine Learning)

College : Government Polytechnic College Washim

Branch : Information Technology

Session : 01

Q1

Q1. Variables & Naming Rules)

Create variables to store the following information about yourself:

Your full name (string)

Your age (integer)

Your height in meters (float)

Whether you are a student (boolean)

Print all four variables. Then explain in comments why you chose those variable names and check their data types using `type()`.

Program Code:

```
# S1Q1.py - Variables & Naming Rules
```

```
# Storing personal details
```

```
full_name = "Vaibhav Vijay Kate"
```

```
age = 17
```

```
height_in_meters = 1.76
```

```
is_student = True
```

```
# Printing values
```

```
print("Name:", full_name)
```

```
print("Age:", age)
```

```
print("Height:", height_in_meters, "meters")
```

```
print("Is Student:", is_student)
```

```
print("\n--- Checking Data Types ---")
```

```
# Checking data types using type()
```

```
print("full_name type:", type(full_name))
```

```
print("age type:", type(age))
```

```
print("height_in_meters type:", type(height_in_meters))
```

```
print("is_student type:", type(is_student))
```

Output Screenshot:

Q2

Q2. Data Types)

Write a Python program that:

- a) Creates one variable of each core data type: str, int, float, and bool.
- b) Prints each variable along with its data type using type().
- c) Try converting the integer to float and float to integer. Print the results and explain the output.

Program Code:

```
# S1Q2.py - Core Data Types & Typecasting
```

```
# Creating variables of core types
```

```
my_string = "Hello AIML"
```

```
my_integer = 15
```

```
my_float = 9.81
```

```
my_boolean = False
```

```
# Printing variables and their types
```

```
print("String:", my_string, "| Type:", type(my_string))
```

```
print("Integer:", my_integer, "| Type:", type(my_integer))
```

```
print("Float:", my_float, "| Type:", type(my_float))
```

```
print("Boolean:", my_boolean, "| Type:", type(my_boolean))
```

```
print("\n--- Type Conversion ---")
```

```
# Converting types
```

```
converted_int_to_float = float(my_integer)
```

```
converted_float_to_int = int(my_float)
```

```
print("Int to Float:", converted_int_to_float)
```

```
print("Float to Int (Truncated):", converted_float_to_int)
```

Output Screenshot:

Q3

Q3. Operators - Arithmetic)

Write a program that takes two numbers $a = 15$ and $b = 4$ and prints the results of:

Addition, Subtraction, Multiplication, Division, Floor Division, Modulo, and Exponentiation.

Add comments explaining the difference between $/$ and $//$ operators with an example.

Program Code:

```
# S1Q3.py - Arithmetic Operators
```

```
a = 15
```

```
b = 4
```

```
# Performing operations
```

```
print("Addition (a + b):", a + b)
```

```
print("Subtraction (a - b):", a - b)
```

```
print("Multiplication (a * b):", a * b)
```

```
print("Division (a / b):", a / b)
```

```
print("Floor Division (a // b):", a // b)
```

```
print("Modulo (a % b):", a % b)
```

```
print("Exponentiation (a ** b):", a ** b)
```

```
# Note: '/' gives floating quotient (e.g. 15 / 4 = 3.75)
```

```
# '//' rounds down division to nearest lower integer (e.g. 15 // 4 = 3)
```

Output Screenshot:

Q4

Q4. Operators - Comparison & Logical)

Given:

x = 25

y = 30

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z = 25

Write expressions using comparison and logical operators to check:

Is x greater than y?

Is x equal to z?

Is x less than or equal to y AND y is not equal to z?

Is x greater than y OR x equal to z?

Print all results with descriptive messages.

Program Code:

S1Q4.py - Comparison & Logical Operators

x = 25

y = 30

z = 25

Evaluating expressions

print("1. Is x greater than y?:", x > y)

print("2. Is x equal to z?:", x == z)

print("3. Is x <= y AND y != z?:", (x <= y) and (y != z))

print("4. Is x > y OR x == z?:", (x > y) or (x == z))

Output Screenshot:

Q5

Q5. Input/Output)

Write a program that asks the user to input:

Their name

Their birth year

Then calculate and print their current age (assume current year is 2026. Use proper formatting and add comments explaining why you need to convert the input.

Program Code:

```
# S1Q5.py - Input/Output & Typecasting

# Getting user inputs
name = input("Enter your name: ")
birth_year_str = input("Enter your birth year: ")

# input() returns string, so we convert birth_year to int for calculation
birth_year = int(birth_year_str)
age = 2026 - birth_year

print(f"\nHello {name}!")
print(f"Your age in 2026 will be {age} years.")
```

Output Screenshot:

Q6

Q6. Combined - Variables, Operators & Input)

Create a simple BMI Calculator:

Take height (in meters) and weight (in kg) as input from the user.

Calculate BMI using the formula: $BMI = weight / (height ** 2)$

Print the BMI rounded to 2 decimal places.

Include appropriate comments.

Program Code:

```
# S1Q6.py - BMI Calculator
```

```
print("--- BMI Calculator ---")
```

```
# Inputs converted to float for decimal values
```

```
weight = float(input("Enter weight (kg): "))
```

```
height = float(input("Enter height (meters): "))
```

```
# BMI formula
```

```
bmi = weight / (height ** 2)
```

```
print(f"\nYour BMI is: {round(bmi, 2)}")
```

```
if bmi < 18.5:
```

```
    print("Category: Underweight")
```

```
elif 18.5 <= bmi < 25:
```

```
    print("Category: Normal weight")
```

```
else:
```

```
    print("Category: Overweight")
```

Output Screenshot:

Q7

Q7. Comments)

Write a small program that calculates the area and perimeter of a rectangle.

Take length and width as input.

Use at least 5 meaningful comments (single-line and one multi-line comment).

Follow good commenting practices (explain why, not what for obvious lines).

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Program Code:

```
# S1Q7.py - Rectangle Area & Perimeter
```

```
"""
```

```
Calculates area and perimeter of a rectangle  
based on user length and width.
```

```
"""
```

```
# Dimensions can have decimals, so we use float
```

```
length = float(input("Enter length: "))
```

```
width = float(input("Enter width: "))
```

```
# Area = length * width
```

```
area = length * width
```

```
# Perimeter = 2 * (length + width)
```

```
perimeter = 2 * (length + width)
```

```
print(f"\nArea: {area} sq units")
```

```
print(f"Perimeter: {perimeter} units")
```

Output Screenshot:

Q8

Q8. Operators - Assignment & Membership)

```
fruits = ["mango", "apple", "banana", "cherry"]
```

```
score = 50
```

Perform the following operations:

Add 25 to score using assignment operator (+=).

Check if "apple" is in the fruits list.

Check if "grape" is not in the list.

Print all results with clear messages.

Program Code:

```
# S1Q8.py - Assignment & Membership Operators
```

```
fruits = ["mango", "apple", "banana", "cherry"]
```

```
score = 50
```

```
# 1. Update score using += shorthand
```

```
score += 25
```

```
print("Updated score:", score)
```

```
# 2. Check if apple is in the list
```

```
print("Is 'apple' in fruits?:", "apple" in fruits)
```

```
# 3. Check if grape is not in the list
```

```
print("Is 'grape' not in fruits?:", "grape" not in fruits)
```

Output Screenshot:

Q9

Q9. Mini Project - All Concepts)

Write a Student Profile Generator program that:

Takes input: name, age, city, and marks in 3 subjects.

Calculates total marks and percentage.

Prints a formatted profile with all information.

Use meaningful variable names, proper comments, and different data types.

Program Code:

```
# S1Q9.py - Student Profile Generator
```

```
print("=== Student Profile Generator ===")
```

```
name = input("Enter name: ")
```

```
age = int(input("Enter age: "))
```

```
city = input("Enter city: ")
```

```
# Marks for three subjects
```

```
sub1 = float(input("Enter Subject 1 marks: "))
```

```
sub2 = float(input("Enter Subject 2 marks: "))
```

```
sub3 = float(input("Enter Subject 3 marks: "))
```

```
# Total and Percentage calculation
```

```
total = sub1 + sub2 + sub3
```

```
percentage = (total / 300) * 100
```

```
print("\n" + "="*30)
```

```
print("  STUDENT PROFILE  ")
```

```
print("="*30)
```

```
print(f"Name   : {name}")
```

```
print(f"Age    : {age} years")
```

```
print(f"City   : {city}")
```

```
print(f"Marks   : {sub1}, {sub2}, {sub3}")
```

```
print(f"Total   : {total}/300")
```

```
print(f"Percentage: {percentage:.2f}%")
```

```
print("="*30)
```

Output Screenshot:

Q10

Q10. Debugging & Understanding)

The following code has some issues. Correct it and add proper comments explaining the fixes:

```
Name = "Alice"
age 20
city = Amsterdam
print("My name is" + Name)
print("I am" age "years old")
print("I live in " city)
# Check if age is greater than 18 and print message
print("Adult:" age > 18)
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```

Program Code:

```
# S1Q10.py - Debugging Python Code

# Fixes made:
# 1. Added '=' to 'age = 20'
# 2. Wrapped "Amsterdam" in double quotes
# 3. Added space in "My name is "
# 4. Added commas in print statements to avoid syntax errors

# Corrected code
Name = "Alice"
age = 20
city = "Amsterdam"

print("My name is " + Name)
print("I am", age, "years old")
print("I live in", city)

# Check if adult (fixed: added comma)
print("Adult:", age > 18)
```

Output Screenshot:
